

Abdullah Al Maruf

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EDUCATION

Bangladesh University of Business & Technology (BUBT), Dhaka, Bangladesh 2018 — 2022
B.Sc. in Computer Science & Engineering (CSE) Cumulative GPA: 3.44/4.00
Thesis Title: Emotion Detection and Prediction of Human Psychology Using Machine Learning Algorithm

EMPLOYMENT HISTORY

Business Automation Limited Mirpur DOHS, Dhaka, Bangladesh
Software Developer-JAVA (Young Professional) 2022 — 2023

- Developed and maintained APIs and backend services for platforms, including projects for the Dutch Bangla Bank and government initiatives.
- Implemented Cypress automation testing, improving software quality assurance processes and reducing manual testing time by almost 20%.
- Enhanced backend systems for e-commerce platforms, including the Gas Monkey website, using Spring Boot frameworks, improving performance, scalability, and maintainability.

VMS CLASS Remote, India
Python Instructor 2023 — Present

- Designed and delivered comprehensive Python coding courses for an online learning platform.
- Conducted regular feedback sessions, addressing challenges and ensuring a student satisfaction rate.
- Created interactive materials and assignments to enhance student engagement and learning outcomes.

Freelance Researcher
Research Assistant 2022 — Present

- Collaborated with international professors on multidisciplinary research projects, contributing to Scopus-indexed publications.
- Demonstrated strong research skills in applying advanced methodologies and analytical techniques to solve complex problems in image and natural language processing, including feature engineering, model building, and hyperparameter tuning

RESEARCH INTERESTS

Machine Learning, Deep Learning, Natural Language Processing (LLM, Generative AI, Text Analysis), Image Processing (Classification, Object Detection, Medical Image Analysis), Cyber-security, Cognitive AI, Explainable Artificial Intelligence

AWARDS AND GRANTS

Research Grants, Bangladesh University of Business & Technology (BUBT) Dhaka, Bangladesh

- Awarded grant to cover Article Processing Charges (APC) for the journal paper, "Emotion Detection from Text and Sentiment Analysis of Ukraine-Russia War Using Machine Learning Techniques." (2022)
- Received grant to support conference participation and publication fees for the paper, "Novel Weighted Ensemble Strategy Using Transfer Learning for Enhanced Cervical Cancer Classification." (2024)

Tuition Fee Waiver, Bangladesh University of Business & Technology (BUBT) Dhaka, Bangladesh, 2020

- Awarded 25% tuition fee waiver in the 8th semester for outstanding academic performance.

PUBLICATIONS

Journal Papers

- **Challenges and Opportunities of Text-Based Emotion Detection: A Survey** — A.A. Maruf, F. Khanam, M.M. Haque, Z.M. Jiyad, M.F. Mridha, and Z. Aung. *IEEE Access*, 2024, Vol. 12, pp. 18416–18450. **Scopus Q1**. DOI: 10.1109/ACCESS.2024.3356357.

- **Classification of Freshwater Fish Diseases in Bangladesh Using a Novel Ensemble Deep Learning Model: Enhancing Accuracy and Interpretability** — A.A. Maruf, S.H. Fahim, R. Bashir, R.A. Romy, S.I. Chowdhury, and Z. Aung. *IEEE Access*, 2024, Vol. 12, pp. 96411–96435. **Scopus Q1**. DOI: 10.1109/ACCESS.2024.3426041.
- **Hate Speech Detection in the Bengali Language: A Comprehensive Survey** — A.A. Maruf, A.J. Abidin, M.M. Haque, Z.M. Jiyad, A. Golder, R. Alubady, and Z. Aung. *Journal of Big Data*, 2024, Vol. 11, No. 1, Article 97. **Scopus Q1**. DOI: 10.1186/s40537-024-00956-z.
- **Emotion Detection from Text and Sentiment Analysis of Ukraine Russia War Using Machine Learning Technique** — A.A. Maruf, Z.M. Ziyad, M.M. Haque, and F. Khanam. *International Journal of Advanced Computer Science and Applications*, 2022, Vol. 13, No. 12. **Scopus Q3**. DOI: 10.14569/IJACSA.2022.01312101.

Conference Proceedings

- **Prediction of Heart Disease and Heart Failure Using Ensemble Machine Learning Models** — A.A. Maruf, A. Golder, A. Al Numan, M.M. Haque, and Z. Aung. *Intelligent Systems*, Springer, 2023, pp. 481–492. DOI: 10.1007/978-981-99-3932-9_41.
- **Ensemble Approach to Classify Spam SMS from Bengali Text** — A.A. Maruf, A. Al Numan, M.M. Haque, T.T. Jidney, and Z. Aung. *International Conference on Advances in Computing and Data Sciences*, Springer, 2023, pp. 440–453. DOI: 10.1007/978-3-031-37940-6_36.
- **Development of Automatic Number Plate Recognition System of Bangladeshi Vehicle Using Object Detection and OCR** — A.A. Maruf, A. Golder, M.S. Naser, A.J. Abidin, A.A.C. Giti, and Z. Aung. *International Conference on Advances in Data-driven Computing and Intelligent Systems*, Springer, 2023, pp. 331–342. DOI: 10.1007/978-981-99-9524-0_5.
- **Covid-19 Vaccine Sentiment Detection and Analysis Using Machine Learning Technique and NLP** — A.A. Maruf, M.N.H. Biplob, and F. Khanam. *International Conference on Machine Intelligence and Emerging Technologies*, Springer, 2022, pp. 401–414. DOI: 10.1007/978-3-031-34619-4_32.
- **A Survey on Personality Prediction** — A.A. Maruf, M.A. Al Nayem, M.M. Haque, Z.M. Jiyad, A.M.O. Rashid, and F. Khanam. *Proceedings of the 2nd International Conference on Computing Advancements*, 2022, pp. 407–414. DOI: 10.1145/3542954.3543012.
- **DDoS Attack Classification Leveraging Data Balancing and Hyperparameter Tuning Approach Using Ensemble Machine Learning with XAI** — Z.M. Jiyad, A.A. Maruf, M.M. Haque, M.S. Gupta, A. Ahad, and Z. Aung. *2024 Third International Conference on Power, Control and Computing Technologies (ICPC2T)*, IEEE, 2024, pp. 569–575. DOI: 10.1109/ICPC2T60072.2024.10475035.
- **Exploring Elderly Activity Recognition Using Deep Learning Through Sensor-Enabled Android App Data Collection** — M.M. Haque, A.A. Maruf, A.J. Obeid, H.K. Judi, M.H. Bhuiyan, and Z. Aung. *2024 Third International Conference on Power, Control and Computing Technologies (ICPC2T)*, IEEE, 2024, pp. 582–587. DOI: 10.1109/ICPC2T60072.2024.10474756.
- **Computer Vision Based Bangla Sign Language Recognition Using Transfer Learning** — M.R. Sadik, R.I. Sony, N.N.I. Prova, A.A. Maruf, S.H. Fahim, and M.S. Islam. *2024 Second International Conference on Data Science and Information System (ICDSIS)*, IEEE, 2024, pp. 1–7. DOI: 10.1109/ICDSIS61070.2024.10594269.
- **Enhancing Crop Management: Ensemble Machine Learning for Real-Time Crop Recommendation System from Sensor Data** — S. Hossain, N.N.I. Prova, M.R. Sadik, and A.A. Maruf. *2024 International Conference on Smart Systems for Applications in Electrical Sciences (ICSSES)*, IEEE, 2024, pp. 1–6. DOI: 10.1109/ICSSES62373.2024.10561406.

For a full list of my publications, please visit: [My Website](#) (Including Pending Publications).

Articles Under Review

- **TransembleNet: Enhancing Species and Vector Mosquito Classification through Transfer Learning Base Ensemble Model** — A.A. Maruf (1st Author), *PLOS ONE*, **Scopus Q1** (Under third revision).
- **EmoBang: Detecting Emotion From Bengali Texts** — A.A. Maruf (1st Author), *SN Computer Science*, **Scopus Q1** (Under first revision).
- **Fostering Trust and Interpretability in Skin Cancer Classification: A Hybrid Framework of Deep Learning and Machine Learning with Explainable AI** — A.A. Maruf (2nd Author), *Heliyon*, **Scopus Q1** (Under first revision).
- **Multi-label Emotion and Sentiment Detection from Bengali Text Using Hybrid Deep Learning Model** — A.A. Maruf (2nd Author), *Computer Speech & Language*, **Scopus Q1** (Under review).

SKILLS

Programming Languages: Python, Java, C++, C, PHP.

Frameworks & APIs: Spring Boot, Spring, Hibernate, REST API, JDBC.

Machine Learning

- **Supervised Learning:** Regression, Classification
- **Unsupervised Learning:** Clustering, Dimensionality Reduction

- **Deep Learning:** Neural Networks, CNNs, RNNs, Transformers
- **NLP:** Word Embeddings, Transformer Models, Text Classification
- **Computer Vision:** Image Classification, Object Detection, Image Segmentation
- **Tools:** TensorFlow, PyTorch, OpenCV, Grad-CAM, SHAP, LIME
- **Explainable AI:** Advanced model transparency and decision interpretability techniques

Data Science

- Data Analysis & Visualization, Statistical Modeling, Data Mining
- Data Cleaning & Preprocessing, Data scraping
- **Databases:** MySQL, Oracle

Web Development: HTML, CSS, JavaScript, Apache Web Server, Tomcat Web Server.

Software Engineering: Software Testing: Cypress automation testing, Version Control: Git.

Tools

- **Development Environments:** IntelliJ, PyCharm, VS Code, Android Studio, Anaconda, Google Colab
- **Productivity Tools:** Microsoft Office, LaTeX, Overleaf

Other Skills

- **Problem Solving:** Active on competitive coding platforms like [LeetCode](#).
- Languages: Fluent in English (B2 level), Bengali

PROJECTS

Transport Management System (C) *Lead Developer*

- Developed a transport management system using C programming to optimize scheduling and vehicle tracking.
- Implemented features for route optimization, booking management, and fleet monitoring.
- Successfully improved operational efficiency by designing robust algorithms for transport logistics.

Covid-19 Bangladesh (Android Application) *Lead Developer*

- Designed and developed an Android application to provide real-time COVID-19 updates specific to Bangladesh.
- Integrated a feature to display a user's location status as red, green, or yellow zones based on their area's risk level.
- Provided statistical information, including country-wise daily updates on new cases, recoveries, and deaths, as well as district-level statistics in Bangladesh.
- Offered medical tips for Covid-19 prevention and care within the app.
- Enabled users to consult doctors through an in-app telemedicine feature.

Campus Connect (Web Application) *Lead Developer*

- Developed a web application that allows students and faculty to connect exclusively within their campus network.
- Implemented features such as messaging and updates to share campus-related news, events, and announcements.
- Ensured secure access by restricting connections to campus members only, enhancing privacy and exclusivity.

Child Education (Android Application) *Lead Developer*

- Designed an engaging and interactive user interface using a variety of graphics and animations specifically tailored to attract young children.
- Developed educational modules within the app to teach children basic concepts such as counting, letter recognition, and vocabulary building through interactive games and activities.

GitHub Repository

[GitHub](#)

RESEARCH PROJECTS

TransembleNet: Enhancing Species and Vector Mosquito Classification through Transfer Learning Base Ensemble Model — 1st Author. *PLOS ONE. Scopus Q1 (Under third revision)*

Problem Statement: Current mosquito vector classification methods struggle with accuracy, computational cost, and imbalanced datasets.

Aim: To enhance classification accuracy in imbalanced datasets using an optimized ensemble model.

Methodology: Developed an ensemble model combining InceptionV3, ResNet-50, and VGG-16 with transfer learning. The approach utilized fine-tuning and robust augmentation techniques to address class imbalance and reduce computational complexity.

Outcomes: Achieved high precision, recall, and F1-scores, significantly improving classification reliability for real-time applications. This contributes to public health by aiding effective disease management.

Project Link: [GitHub Repository](#)

Multi-label Emotion and Sentiment Detection from Bengali Text Using Hybrid Deep Learning Model — 2nd

Author. *Computer Speech & Language. Scopus Q1 (Under review)*

Problem Statement: Limited datasets and accurate models hinder multi-label emotion and sentiment detection in Bengali.

Aim: To develop an annotated dataset and a hybrid deep-learning model combining CNN, LSTM, and BERT embeddings.

Methodology: Created a dataset of 40,911 Bengali posts with high annotation quality (Cohen's Kappa = 0.85). Employed CNN for local pattern recognition, LSTM for temporal dependencies, and BERT embeddings for feature extraction.

Outcomes: Achieved 99% accuracy and precision, setting a benchmark for Bengali NLP. Introduced a publicly available annotated dataset for future research.

Project Link: [Google Colab](#)

For a complete list of my research projects, please visit: [My Website](#).

TALKS & PRESENTATIONS

Virtual Presentations:

- **7th International Conference on Advances in Computing and Data Sciences (ICACDS 2023).** Kolkata, India: Presented our research paper "Ensemble Approach to classify Spam SMS from Bengali text."
- **3rd International Conference on Machine Learning, IoT and Big Data (ICMIB-2023).** Sarang, India: Presented our research paper, "Prediction of Heart Disease and Heart Failure Using Ensemble Machine Learning."
- **2nd International Conference on Data Science and Information System (ICDSIS 2024):** Presented our research paper, "Computer Vision Based Bangla Sign Language Recognition Using Transfer Learning."

Physical Presentations:

- **2nd International Conference on Machine Intelligence and Emerging Technologies (MIET 2022).** Noakhali, Bangladesh: Presented our research paper, "Covid-19 Vaccine Sentiment Detection and Analysis Using Machine Learning Technique and NLP."
- **27th International Conference on Computer and Information Technology (ICCIT 2023):** Presented our research paper, "Predictive Analysis of Bangladeshi Students' Academic Performances Using Ensemble Machine Learning with Explainable AI Techniques."
- **3rd International Conference on Computing Advancements (ICCA 2023):** Presented our research paper, "A Novel Weighted Ensemble Strategy Using Transfer Learning for Enhanced Cervical Cancer Classification."

For a full list of my talks, please visit: [My Website](#).

OTHER EXPERIENCES

Peer Reviewer for PLOS ONE

2024

- Reviewed text classification, NLP, deep learning, and machine learning manuscripts.
- Provided detailed feedback to authors to enhance the quality of their submissions.
- Offered recommendations to editors on manuscript acceptance or revision.

Peer Reviewer for AIHC, Springer

2024

- Reviewed manuscripts on human activity recognition, sensor data analysis, and feature selection.
- Provided constructive feedback to authors, focusing on methodological and analytical improvements.
- Assisted editors in evaluating the suitability of manuscripts for publication.

Organizer, Departmental Rag Day

Dhaka, Bangladesh

Event Organizer

2022

- Planned and executed a fundraising event, fostering teamwork and community spirit.
- Coordinated with stakeholders to ensure event success and engagement.
- Managed budgeting, scheduling, and logistics for the event.

Speaker, Webinar: Writing Effective Research Papers

Virtual

Speaker

2024

- Delivered a virtual presentation to students on "Writing Effective Research Papers: A Step-by-Step Guide."
- Provided valuable insights on research methodology and academic writing.
- Engaged attendees with practical tips and answered queries during a live Q&A session.

REFERENCES

Zeyar Aung

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Abu Dhabi, UAE

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Google Scholar: [Google Scholar](#)

Fahima Khanam

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